DNA Ligation Using T4 DNA Ligase

M13 | ver 1.1 | Revised 2026-10-03

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Purpose

Join a prepared DNA insert to a linearized plasmid vector using Takara T4 DNA Ligase, then recover and verify recombinant clones. This protocol covers conventional cohesive-end and blunt-end ligation with separately supplied enzyme and 10× buffer. [1, 2]

Materials & Equipment

Item	Supplier / identifier	Specification / use
T4 DNA Ligase	Takara / 2011A or 2011B	350 U/μL; conventional enzyme
10× T4 DNA Ligase Buffer	Takara / supplied with enzyme	ATP-containing buffer; use at 1×
Linearized vector and insert DNA	Prepared samples	Compatible ends; known lengths and concentrations
DEPC-treated water	Sangon Biotech / B501005	Laboratory water for reaction preparation [6]
Tubes and tips	Axygen tips; laboratory tubes	Reaction setup
Pipettes, ice rack, microcentrifuge	Eppendorf centrifuge; laboratory pipettes and ice rack	Dispensing and brief collection
Thermal cycler or temperature-controlled block	Laboratory equipment	Stable 16°C incubation
Competent cells and selection plates	See M05	Transformation; antibiotic matched to vector

Before You Start

Check that the vial is conventional T4 DNA Ligase with a separate 10× buffer. A DNA Ligation Kit or Mighty Mix requires its own formulation and instructions.

Prepare restriction-digested DNA using M03 and purify as needed using M02. Check the intended junctions, end compatibility and fragment sizes. Measure DNA after purification.

Ligation requires a 5′ phosphate and a 3′ hydroxyl at the bond being sealed. With a dephosphorylated vector, provide a phosphorylated insert. Undigested PCR products and synthetic oligonucleotides may lack 5′ phosphates; arrange phosphorylation when needed and anneal

complementary oligos before use. [2, 3]

Remove or appropriately inactivate upstream phosphatase before ligation. Keep enzyme stocks at -20°C between uses. Thaw and mix the buffer completely, briefly spin, and keep the enzyme on ice during setup. [2, 3]

Procedure

A. Calculate DNA inputs

1. Use 50 ng vector and an insert:vector molar ratio of 3:1 as the initial manufacturer example. Optimize within 1:1 to 10:1 when end type, fragment size or vector dephosphorylation warrants it. [1]

Insert mass (ng) = vector mass (ng) × insert length (bp) / vector length (bp) × desired insert:vector molar ratio. For each DNA, volume (µL) = mass (ng) / concentration (ng/µL).

Calculation example only: a 5,000 bp vector (50 ng) and a 1,000 bp insert at 3:1 require 30 ng insert. Use the purified backbone length, not the length of the original uncut plasmid.

B. Assemble the reaction

2. Combine water, buffer and DNA on ice using the following 20 µL reference reaction; add ligase last. [1]

Component	Volume / amount	Final condition
10× T4 DNA Ligase Buffer	2 µL	1×
Linearized vector	50 ng; calculated volume	Reference DNA input
Insert	Calculated volume	3:1 molar ratio initially
T4 DNA Ligase (350 U/µL)	1 µL	350 U per reaction
Nuclease-free water	To 20 µL	17 µL minus total DNA volume
Total	20 µL	

If combined DNA volume exceeds 17 µL, concentrate and remeasure the DNA. For an impractically small transfer, prepare a measured working dilution and use its concentration in the calculation.

3. Prepare a vector-only control with the same vector input, buffer and ligase; replace insert volume with water. When investigating background, also prepare vector without insert or ligase, replacing both with water. [4]

4. Mix gently by pipetting and briefly centrifuge to collect the reaction at the bottom of the tube.

C. Incubate and recover clones

5. Incubate according to DNA end type. Use the shorter end of the range first for a new cohesive-end reaction; extend within the stated range if recovery is poor after checking controls. [1]

End type	Temperature	Time
Compatible cohesive ends	16°C	1-5 h
Blunt ends	16°C	1-24 h

6. Use an aliquot for transformation following M05, preferably its conventional route for a new ligation. At the M05 reference scale, use 1-5 µL reaction per 50 µL chemically competent cells. Include a known plasmid control and a no-DNA control when establishing or troubleshooting the workflow. [5]

Takara's standard workflow proceeds directly to transformation without a specified heat-inactivation step. For a pause, store the reaction at -20°C. Remove salts before electroporation using a compatible DNA-cleanup method. [1, 3]

7. Compare sample and control plates. Pick isolated colonies according to M06 and verify the insert, its orientation and both junctions by sequencing; use colony PCR or diagnostic digestion as preliminary screens where useful. [5]

p.s.

Use molar ratios rather than equal DNA masses. A short annealed oligo can require very little mass; dilute it to a pipettable concentration instead of adding an arbitrary volume.

Use the matching ATP-containing buffer. If a previously reliable ligation fails, try a fresh buffer aliquot before increasing enzyme input; ATP deterioration can limit activity. [3]

After vector dephosphorylation, confirm insert phosphorylation and removal of active phosphatase. Leaving both partners without a usable 5′ phosphate prevents the required joining reaction. [3]

Expected Outcome

Successful ligation yields transformants from which sequence-verified recombinant clones can be recovered. More colonies than the vector-only control supports enrichment but does not prove correct assembly. Colony count, gel migration or a positive colony PCR alone cannot establish the complete intended sequence. Retain sample identities, reaction inputs, control observations and verification results.

Troubleshooting

Observed problem	Check first	Suggested action
Few or no colonies, including plasmid control	Cell competence, selection and transformation handling	Troubleshoot M05 first; a failed transformation control cannot diagnose the ligation.
Few or no colonies, but plasmid control works	End compatibility, 5′ phosphates, buffer condition and DNA purity	Correct end preparation; use fresh buffer and clean DNA. If those checks pass, extend within the applicable incubation range. [1, 3]
Many colonies in vector-only control	Compare vector with and without ligase	High no-ligase background suggests uncut plasmid: revisit M03. Background mainly with ligase suggests vector reclosure; consider dephosphorylation for compatible vector ends. [4]
Insert-containing clones have wrong orientation or junctions	End design, expected construct map and sequence data	For nondirectional ends, screen both orientations. Compare both junctions with the intended sequence and select a verified clone.
Electroporation arcs or gives poor recovery	Salt carryover and cell/electroporator requirements	Desalt the completed reaction before retrying with an appropriate aliquot; follow the cell-specific electroporation procedure. [3]

Reference

[1] Takara Bio. T4 DNA Ligase: ligation of foreign DNA into plasmid vectors, product 2011A/B; procedure and footnotes. Accessed 2026-10-03. Open source

[2] Takara Bio. T4 DNA Ligase, 2011A, product data sheet v201312Da, p. 1: description, storage and supplied buffer. Accessed 2026-10-03. Open source

[3] New England Biolabs. Causes of ligation reaction failure leading to transformation failure. General biochemical and cleanup guidance only; Takara reaction parameters are taken from [1]. Accessed 2026-10-03. Open source

[4] New England Biolabs. Cloning Problem: Too much background in the transformation step. Control interpretation and phosphatase removal. Accessed 2026-10-03. Open source

[5] SUSTech-BIO 2026 iGEM Team. Maintained protocol library: M02 (DNA purification), M03 (digestion), M05 v1.0 (transformation) and M06 (colony handling and verification). Read 2026-10-03.

[6] Sangon Biotech. B501005 DEPC-treated water, SDS, revision 2024-12-24; product identity. Accessed 2026-10-03. Open source